

Interim-1 Report: Fraud Detection Data Analysis and Preprocessing

Project: Improved Detection of Fraud Cases for E-commerce and Bank Transactions

Prepared for: Adey Innovations Inc. challenge

Submission focus. This interim report covers Task 1: data cleaning, exploratory analysis, geolocation integration, feature engineering, transformations, and imbalance handling. The raw CSV datasets were not included with the uploaded brief, so this package includes runnable scripts that produce the numeric tables and plots after the datasets are placed in `data/raw/`.

1. Business framing

Fraud detection must balance false positives, which block legitimate customers, and false negatives, which allow direct financial loss. The project therefore treats accuracy as secondary and prepares class-aware outputs such as class imbalance ratios, AUC-PR, F1-score, and confusion matrices for later modeling.

The e-commerce and credit card streams are modeled separately because their feature structures differ. E-commerce has user, device, browser, timestamp, and IP context. Credit card data contains anonymized PCA features plus time and amount.

2. Data sources and target fields

Dataset	Target	Main preprocessing needs
Fraud_Data.csv	class	Timestamp parsing; duplicate removal; IP-to-country merge; categorical encoding; temporal and velocity features.
IpAddress_to_Country.csv	-	IP range cleaning and range-based lookup.
creditcard.csv	Class	Duplicate removal; amount/time features; scaling; imbalance handling.

3. Data cleaning approach

The code removes exact duplicate rows before feature engineering. Dates are converted with safe coercion, numeric fields are preserved as numeric, and missing numeric values are imputed with medians during model preprocessing. Categorical missing values are imputed with the most frequent value. Country values that cannot be matched through IP ranges are assigned Unknown rather than dropped.

Dropping rows is avoided unless a row is structurally unusable because fraud examples are rare and every positive case is valuable.

4. Geolocation integration

The e-commerce IP address is converted to an integer using IPv4 arithmetic. The IP lookup table is sorted by lower bound, then a range-aware `merge_asof` operation attaches the nearest lower IP range. A match is retained only when the transaction IP is also less than or equal to the range upper bound. Otherwise, country is set to Unknown.

The EDA code also produces fraud rate by country, filtered to countries with enough transactions to avoid overreacting to tiny sample sizes.

5. Feature engineering choices

Feature	Dataset	Reason
hour_of_day	Both	Captures time-of-day fraud patterns and unusual transaction windows.
day_of_week	Fraud_Data.csv	Captures weekly behavioral cycles.
time_since_signup_hours	Fraud_Data.csv	Flags very fast purchases after signup, a common account-abuse risk signal.

Feature	Dataset	Reason
user_transaction_count	Fraud_Data.csv	Simple transaction-frequency proxy.
device_transaction_count	Fraud_Data.csv	Highlights shared devices across many users.
purchase_value_log1p / amount_log1p	Both	Reduces skew in monetary values while preserving ordering.
country	Fraud_Data.csv	Adds geolocation risk signal from IP ranges.

6. EDA and plot outputs

The Task 1 script generates class distribution plots for both datasets, purchase/amount by class plots, and top-country fraud-rate plots under `reports/figures/`. These plots are ready to insert into the GitHub report or notebooks after the real CSV files are added.

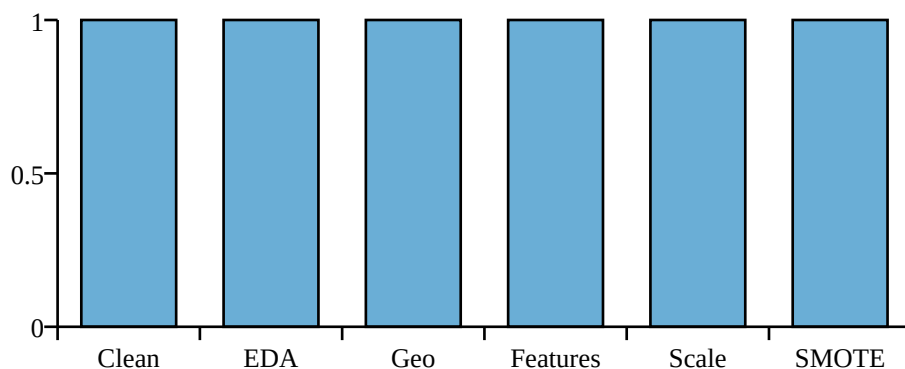


Figure 1. Interim-1 deliverable coverage implemented in the repository. Actual dataset-specific plots are generated after raw CSV files are added.

7. Data transformation and imbalance handling

Numerical variables are imputed with the median and scaled with `StandardScaler`. Categorical variables are imputed with the most frequent category and one-hot encoded with unknown-category handling enabled. The split is stratified to preserve fraud prevalence in train and test sets.

SMOTE is applied only to the training set after the split. This avoids synthetic examples leaking into validation or test data. The script writes `data/processed/resampling_summary.json` with class counts before SMOTE, after SMOTE, and in the unchanged test set.

8. Interim-1 deliverables checklist

Requirement	Included in package
Cleaned datasets	Generated as Parquet files by <code>scripts/run_task1.py</code> .
EDA report and visualizations	PDF report plus automated plots in <code>reports/figures/</code> .
Feature engineering documentation	README, report, and <code>src/features.py</code> .
IP-to-country mapping	Implemented in <code>src/features.py</code> using <code>add_country_by_ip</code> .
Class imbalance strategy	SMOTE-on-training-only implementation and written justification.
Repository structure	Matches requested folders and includes CI test workflow.

9. Limitation and next step

The uploaded materials included the task brief but not the three raw CSV datasets. The report therefore cannot truthfully state actual class percentages, country fraud rates, or data-specific outliers. After adding the raw files, run `python scripts/run_task1.py` to populate all numeric outputs and plots, then run the Task 2 and Task 3 scripts for model metrics and SHAP interpretation.